

Lecture 3 — Eigenvalues and Eigenvectors

1. Motivation

Many linear transformations have privileged directions: vectors that the map merely stretches. Understanding these directions reduces a complicated matrix to a handful of scalars, and underpins diagonalization, stability analysis, and spectral methods.

2. Definitions

Let A be an $n \times n$ matrix. A nonzero vector v is an eigenvector of A with eigenvalue λ when:

$$A v = \lambda v$$

3. The characteristic polynomial

Eigenvalues are exactly the roots of the characteristic polynomial, obtained by requiring that $A - \lambda I$ be singular:

$$\det(A - \lambda I) = 0$$

4. Worked example

For the 2×2 shear-and-stretch matrix discussed in class, the characteristic polynomial factors as $(\lambda - 2)(\lambda - 1/2)$, giving eigenvalues 2 and $1/2$ with independent eigenvectors v_1 and v_2 . See Figure 4.2 in the lecture notes.

5. Exercises

Compute the spectrum of the rotation matrix $R(\theta)$ and explain geometrically why no real eigenvectors exist for $\theta \notin \{0, \pi\}$. Then show that symmetric matrices always admit an orthonormal eigenbasis.